# PS 06 — RoutePulse: Optimise Campus Transport Under Real Constraints

Constraints

## Official problem statement

Large campuses and institutional transport systems may operate multiple routes, fixed departure windows, capacity-limited vehicles, and stops with uneven passenger demand. Students need reliable arrival information and feasible connections, while transport operators must allocate vehicles and schedules efficiently. A system that only displays vehicle locations does not resolve the underlying planning problem. Develop a transport-planning system that estimates arrival windows, forecasts passenger demand, and recommends feasible route or vehicle adjustments while respecting vehicle capacity, timetable, travel- time, and service constraints. The solution should demonstrate an improvement against a defined baseline using common simulated transport data.

## Mandatory MVP

• Represent routes, stops, schedules, vehicle capacities, and passenger demand in a usable data

model or interface.

• Estimate arrival windows from supplied or simulated GPS traces and state relevant assumptions.

• Identify likely overcrowding, capacity violations, missed connections, or schedule conflicts.

• Recommend at least one feasible route, schedule, or vehicle-allocation adjustment and explain the

trade-off.

• Compare the proposed plan against a documented baseline schedule using measurable outcomes.

## Optional extensions

• Forecast demand by stop, time window, weekday, or event schedule.

• Optimize vehicle allocation or route adjustments under explicit constraints.

• Model disruptions such as traffic delays, vehicle unavailability, or sudden demand spikes.

• Provide passenger-facing alternatives or accessibility-aware routing.

• Integrate a map, live location feed, or an edge-based passenger-counting prototype; real hardware

and live feeds are optional.

## Data and feasibility

Organizers should supply synthetic GPS traces, stop coordinates, route definitions, schedules, vehicle capacities, passenger counts, and scenario constraints. Include normal operation and a small number of disruption scenarios. The baseline and evaluation time windows must be fixed before judging. Actual campus fleet integration is an optional extension.

## Commercial model

Campus transport planning software, fleet-management subscriptions, and licensing to private universities, institutional transport operators, and fleet contractors.